

METHODOLOGY

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Scalable unsupervised labeling with SHAP feature selection for fraud detection in imbalanced data

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Abstract

There is a growing need for labeled data, yet manual annotation is costly, error-prone, and often infeasible in privacy-sensitive, highly imbalanced domains such as fraud detection. We introduce a fully unsupervised framework that combines unsupervised SHapley Additive exPlanations (SHAP) feature selection with our novel unsupervised labeling method. We apply unsupervised SHAP to the Kaggle Credit Card Fraud Detection and Medicare Part D datasets to produce high-impact feature subsets, and then label the datasets with our unsupervised labeling approach. To effectively evaluate the labels generated by our novel methodology, we apply a baseline unsupervised learner, Isolation Forest (IF), to both the original datasets and their subsets. We calculate Matthew's Correlation Coefficient (MCC), Jaccard Index (JI), Precision, Recall, and F1-score by comparing our generated labels against the ground truth labels. It is important to note, the ground truth labels were used solely for evaluation. Our empirical results surpass the results obtained with the full feature dataset and baseline. By improving label quality while reducing computational complexity and preserving privacy, our approach offers a practical solution for learning from unlabeled, severely imbalanced data.

Keywords: Label generation, Unsupervised labeling, Big Medicare fraud data, Credit card fraud detection, Class imbalance, SHAP

Introduction

A fundamental challenge in machine learning is acquiring high-quality labels, as manual labeling is costly, time-consuming, and prone to errors [1]. The performance and generalizability of machine learning models are oftentimes dependent on label quality; therefore, the absence of labeled data limits the applicability of supervised learning approaches, particularly in domains such as fraud detection and medical diagnostics [2]. Another complication is privacy concerns given that data annotation often requires an expert to be given access to sensitive information, restricting the availability of labeled datasets further [3].

Fraud detection presents additional challenges due to severe class imbalance, where fraudulent transactions are vastly outnumbered by legitimate ones. Often, literature

focuses on moderate class imbalances (1:4 to 1:100); however, real-world fraud datasets typically exhibit severe imbalance (1:1000 to 1:5000) [4]. The datasets analyzed in this study, Kaggle's Credit Card Fraud Detection (1:577) [5] and Medicare Part D (1:1445) [6], are real-world fraud datasets with severe class imbalance. Existing unsupervised labeling methods struggle with this level of class imbalance and often fail to generate quality labels leading to high false positive rates (legitimate instances labeled as fraudulent), which results in wasted time and financial resources investigating cases that are not fraudulent.

To overcome these issues, we present our novel unsupervised labeling methodology that improves label quality, even when severe class imbalance is present. Our approach combines two labeling strategies, an Ensemble Unsupervised Method (EUM) and a Percentile Gradient Method (PGM) to generate initial labels, which is followed by a minimization process fine-tune positive instances and reduce false positives. We take the instances present in the intersection of minimized EUM (mEUM) and minimized PGM (mPGM) to produce a subset of the original dataset, the confident labels (CLs), which are then used to build confidence intervals. The CLs are then used to select the final positive instances. Our method provides a scalable solution applicable to both balanced and highly imbalanced datasets.

Another challenge in generating labels in machine learning is the size of the dataset [7]. Unsupervised and supervised methods can be computationally expensive especially when applied to large and big data. Because of this, feature selection has gained attention as a method to manage the computational expense by limiting the number of features that need to be considered [8]. Feature selection can also play a beneficial role when in datasets have class imbalance, as feature selection aims to identify and retain the most relevant features, which allows for better interpretability [8]. Our work integrates feature selection in combination with our novel unsupervised labeling technique to address this variety of challenges. Our empirical results show the effectiveness of feature selection to further improve the quality of labels generated by our unsupervised labeling method.

We integrate SHapley Additive exPlanations (SHAP) [9] to perform unsupervised feature selection, reducing computational complexity while enhancing label quality. Unlike traditional supervised feature selection, SHAP enables the identification of important features without relying on labeled data. We apply unsupervised SHAP-based feature selection to extract the top features for each dataset. For the Credit Card Fraud Detection dataset, three and five features were selected, whereas for the Medicare Part D dataset, we selected three and seven features. We selected these top feature combinations for each dataset after evaluating multiple levels of feature selection, where the selected feature subsets consistently had the highest label quality. Additionally, we include the full-featured version of each dataset. This resulted in a total of six datasets, four with feature selection and two without. We compare the labels of the data subsets to their corresponding full-featured datasets, and our results demonstrate that unsupervised SHAP-based feature selection improves our unsupervised labeling method, achieving a greater proportion of true positives while reducing the number of false positives.

To assess our label quality, we compare our method against both the full feature set and an unsupervised baseline. Lacking a fully automated labeling approach in the literature, we adopt Isolation Forest (IF) [10]—a scalable anomaly detector that

outperforms Gaussian Mixture Models and k-NN [11, 12] while avoiding the instability of clustering [13]—as our baseline. We apply IF to all six datasets and compare its labels with those generated by our unsupervised labeling method. Across each evaluation metric, SHAP consistently yields higher scores than both IF and the full feature set, demonstrating improved label quality with reduced computational cost.

Fraud detection is an anomaly detection task, requiring models to identify rare fraudulent patterns among a majority of legitimate transactions. Fraud takes various forms, including credit card fraud, healthcare fraud, and identity theft, all of which impose severe financial and emotional burdens on victims [14, 15]. In the United States, credit card fraud affects 60% of cardholders, with annual losses exceeding \$5 billion [16]. The increasing complexity of fraud schemes furthers the need for machine learning-driven detection, which can reduce financial losses, streamline investigations, and enhance consumer protection. Fraud is a widespread issue, affecting individuals, businesses, and governments [17]. Advancements in digital technology have led to its growing prevalence, with 93% of credit card fraud cases involving remotely accessed account data rather than physical card theft [16]. The Federal Trade Commission reports that fraud losses exceeded \$10 billion in 2023, the highest on record [18]. The financial impact of fraud is

staggering. Identity theft resulted in \$16.4 billion in losses in 2021 [15], while Medicare fraud costs \$60 billion annually [19]. Government losses are equally disturbing, with improper payments totaling \$2.7 trillion since 2003 and taxpayer losses ranging from \$233 to \$521 billion annually [20]. Despite recovery efforts, such as the Department of Justice reclaiming \$3 billion in 2019, fraud remains a significant financial threat [21].

This paper makes four primary contributions to unsupervised fraud detection labeling. First, it refines our earlier, partially supervised method [22] into a fully unsupervised, end-to-end framework that labels every instance and is appropriate for severely imbalanced, big data by combining an ensemble of unsupervised learners with percentile thresholding. Our improved approach [23] removes any reliance on domain expertise or downstream classifiers. Second, our approach is combined with unsupervised SHAP-based feature selection, which simultaneously reduces computational load and improves label quality. Third, we introduce a direct label quality evaluation, using MCC, Jaccard, Precision, and Recall. Fourth, a comprehensive study on the Kaggle Credit Card Fraud and Medicare Part D datasets shows that our approach consistently surpasses an unsupervised baseline, Isolation Forest baseline, both with and without feature selection, and details how varying positive class instances affects performance. Collectively, these contributions advance automated labeling techniques, providing a scalable and efficient solution for machine learning applications in fraud detection and other domains.

The remainder of this paper is organized as follows. “[Related works](#)” section highlights current methods of unsupervised labeling and highlights the novelty of our work in these domains. “[Methodology](#)” section offers an in-depth explanation of our methodology. In “[Experimental design](#)” section, we detail the datasets used, the method of label evaluation, performance metrics, and baseline unsupervised algorithm for comparison to our method. “[Results and analysis](#)” section presents our empirical results and analysis. Finally, “[Conclusion](#)” section concludes the paper and discusses possible areas of future work and application related to our method.

Related works

The literature presented explores the application of SHapley Additive exPlanations (SHAP) in feature selection, class label generation and evaluation, particularly in fraud detection tasks involving imbalanced and large datasets. SHAP has been integrated into various anomaly detection and classification methodologies, demonstrating its utility in both supervised and unsupervised settings, as shown by Hancock et al. [24]. While SHAP has been widely used for data reduction and model performance improvement, its role within fully unsupervised frameworks remains an emerging area of interest. Many existing approaches incorporate pseudo-labeling, supervised models, or ensemble methods, making them partially dependent on labeled data.

Park et al. [25] introduce a SHAP-driven pseudo-labeling method for text data augmentation using explainable artificial intelligence (XAI). Their approach assigns word importance scores via SHAP and refines pseudo-labels by combining them with ground-truth labels during training. While our method also employs SHAP to improve labeling, it differs in that it is fully unsupervised. Park et al. require a pre-trained model for SHAP calculations and incorporate pseudo-labels alongside original labels, whereas our method generates labels solely from our unsupervised labeling method with features minimized by SHAP feature selection and requiring no prior labels or model training.

Liu et al. [26] use SHAP-based feature selection on the UCI Parkinson's disease dataset, integrating SHAP values with Deep Forest, Extreme Gradient Boosting, LightGBM, and Random Forest. Their approach compares SHAP-based selection against filter-based methods such as F-score, ANOVA-F, and Mutual Information. However, their method depends on supervised models to compute SHAP values and applies feature selection only to the training dataset. In contrast, our approach applies SHAP in a fully unsupervised manner across the entire dataset before labeling.

Ghaheiri et al. [27] propose a machine learning-based approach for early Parkinson's disease (PD) diagnosis using voice recordings. Their method selects features using Pearson Correlation Coefficients (PCCs), ranks them with SHAP, and then classifies data using XGBoost, LightGBM, Gradient Boosting, and Bagging. Predictions are determined through a weighted Hard Voting Ensemble. However, their approach relies on labeled data and supervised models, limiting its applicability to fully unlabeled or imbalanced datasets. In contrast, our method is entirely unsupervised, applying unsupervised SHAP-based feature selection before any labeling. By ranking and selecting features prior to labeling, our approach reduces computational complexity and enhances scalability for large, imbalanced datasets. Unlike Ghaheiri et al., our method generates labels in an unsupervised framework, making it more suitable when labeled data is scarce.

Various machine learning models, including Multilayer Perceptrons (MLP), Convolutional Neural Networks (CNN), and Support Vector Machines (SVM), have been applied for early diagnosis, Medicare fraud detection, credit card fraud detection, product defect identification, and machine fault detection [28–31]. However, a fundamental limitation of supervised models is their dependence on labeled data, which restricts their applicability to domains where labeled instances are scarce or expensive to obtain. Manual labeling is time-consuming, costly, and requires domain expertise, posing a persistent challenge for machine learning practitioners [32]. In contrast, the advancement of unsupervised methods, such as the one proposed in this study, addresses this limitation by

eliminating the need for labeled data while still enabling effective fraud detection. Our approach leverages SHAP for feature selection to enhance label quality, making it a scalable and practical solution for large, imbalanced, and unlabeled datasets.

Caracillo et al. [33] propose a hybrid fraud detection approach that integrates unsupervised and supervised techniques. They compute outlier scores using methods such as z-score, Principal Component Analysis (PCA), and Isolation Forest (IF) to identify anomalies without labels. However, these outlier scores are then incorporated as features into a supervised model, specifically Balanced Random Forest (BRF), which is trained on labeled data to detect fraudulent activity. While their method leverages unsupervised techniques to assess anomalies, it ultimately depends on labeled data for final fraud detection. In contrast, our approach remains fully unsupervised, identifying and labeling instances without relying on pre-existing labels, making it more suitable for scenarios where labeled data is unavailable or costly to obtain.

Jiang et al. [34] present an anomaly detection framework (UAAD-FDNet) for credit card fraud detection, training their model exclusively on non-fraudulent transactions while treating fraud as an anomaly. Their network consists of a generator and discriminator, where the generator uses an autoencoder with feature attention to reconstruct features and learn latent representations. While they use the Kaggle Credit Card Fraud Detection dataset, their data preparation requires labels to isolate non-fraudulent transactions, making their method partially supervised. In contrast, our approach is entirely unsupervised, classifying both fraudulent and non-fraudulent transactions without pre-defined labels. Additionally, Jiang et al. use feature attention, whereas we use feature selection, which retains all instances but reduces complexity by selecting only the most important features.

Moslehi et al. [35] introduce a semi-supervised cluster labeling approach that employs iterative K-means clustering to refine clusters and identify influential features. Their method assigns labels by matching cluster statistics from the dataset to a secondary, pre-labeled dataset, effectively using it as a reference for label assignment. In contrast, our approach is fully unsupervised, as it does not rely on an external labeled dataset to determine final labels. While K-means is incorporated as part of our ensemble labeling strategy, our method assigns labels independently, without the need for prior labeled information. Additionally, their study is limited to a small dataset of 385 samples, whereas our method is successfully applied to two large-scale datasets containing approximately 285,000 and 5.3 million instances, demonstrating its scalability and effectiveness in handling big data.

Babu et al. [36] and Naby et al. [31] utilize supervised learning approaches on the Kaggle Credit Card Fraud dataset to label unseen transactions. Babu et al. employ a convolutional neural network (CNN), which achieves strong performance but requires ground truth labels during training, a limitation inherent to supervised methods. Similarly, Naby et al. investigate techniques to improve model training for label prediction using supervised learning. In contrast, our approach is fully unsupervised, generating binary labels without relying on original labels. Another key distinction is the choice of evaluation metrics. Both studies rely on accuracy, which is known to be misleading in imbalanced datasets like the Credit Card Fraud dataset [37]. While Naby et al. incorporate precision and recall, their primary reliance on accuracy remains problematic. Instead, we evaluate

label quality using Precision, Recall, Matthews Correlation Coefficient (MCC), and Jaccard Index, which are well-suited for imbalanced data and have been demonstrated as robust evaluation metrics [38, 39]. Our direct label evaluation approach ensures that the generated labels are reliable and effective, avoiding the biases introduced by supervised learning and inappropriate performance metrics.

Gadetsky et al. [40] present HUME, a model-agnostic framework which infers human labeling in an entirely unsupervised manner. Utilizing diverse pre-trained representations (self-supervised, supervised, and weak-supervision models), they demonstrate state-of-the-art (SOTA) performance on multiple large-scale benchmarks aiming to find consistent labels which align with human annotations. In contrast, our labeling method does not search for labels or require pre-trained models, rather, it directly generates binary labels through an entirely unsupervised framework, thereby streamlining the labeling process.

Recent literature [41] highlighting various state-of-the-art unsupervised methods, including One-Class SVM (OCSVM), Isolation Forest (IF), Local Outlier Factor (LOF), and Robust Covariance (RC), providing a comprehensive evaluation of their predictive performance based on accuracy, precision, recall, and F1-score. The results indicate that OCSVM, IF, and RC are the most effective in detecting outliers, with IF demonstrating a slight advantage by achieving a better balance between precision and recall. Additionally, the study emphasizes IF's efficiency and effectiveness in handling high-dimensional, large-scale datasets. Consequently, we adopt IF as an unsupervised SOTA baseline method for comparison with our unsupervised labeling approach, which is well-suited for large-scale datasets.

The following references highlight existing approaches to label evaluation, providing a comparison to our method. While binary class labels can be generated using various techniques, it is essential to assess their effectiveness. Zhang et al. [42] used weak supervision to generate binary labels and evaluated their quality by training end models and measuring performance with ROC-AUC and PR-AUC scores. Similarly, Shi et al. [43] applied a semi-supervised feature selection method, assessing the reliability of pseudo-labels by training classifiers on selected features and using classification accuracy as the evaluation metric. Another semi-supervised approach [29] generated labels and assessed them through a Support Vector Machine (SVM) model using 10-fold cross-validation, relying on precision, recall, and F-measure for evaluation. A commonality among these methods is that label evaluation is performed indirectly by training a classifier on the generated labels and assessing its performance. In contrast, our approach directly evaluates label quality without the need for a supervised model, making the process more efficient and model-agnostic. This eliminates the complexity of selecting an appropriate classification model [44], as performance can vary significantly depending on the dataset, and ensures a more consistent and unbiased assessment of label quality.

The related works presented differ from our approach in several key ways. First, while many studies use SHAP for feature selection or labeling [25, 26], they incorporate supervised components, such as pretrained models or original labels, requiring labeled data. In contrast, our method applies SHAP in a fully unsupervised manner and generates labels without any reliance on pre-labeled data. Second, some methods require partially supervised data preparation, such as isolating non-fraudulent transactions before

analysis [34]. Our approach does not require such preprocessing; instead, it processes both fraudulent and non-fraudulent transactions and determines a label of legitimate or fraudulent. Third, studies such as [36] and [31] rely on accuracy as their primary evaluation metric, which is less reliable for imbalanced datasets [37]. Instead, we evaluate label quality using Precision, Recall, Matthews Correlation Coefficient (MCC), and Jaccard Index, which are better suited for imbalanced data [39]. Fourth, [34] incorporates feature attention within an unsupervised framework to assign greater weight to significant features. In contrast, we use SHAP for feature selection, which not only identifies and retains the most relevant features but also reduces computational complexity, making our method more scalable and efficient for large, imbalanced datasets. Beyond these distinctions, many related works introduce labeled data at various stages [33, 36], limiting their applicability in fully unsupervised settings. Some methods, such as [35], also may lack scalability, as they are tested on small datasets (e.g., 385 samples), whereas our approach is applied to datasets with up to 5.3 million instances. Additionally, prior studies often evaluate generated labels indirectly by training a supervised classifier [29, 42, 43], adding unnecessary complexity, whereas our method directly assesses label quality, eliminating the need for classifier training. Finally, our approach is fully automated, making it practical for big data tasks, particularly those involving severe class imbalance, unlike prior methods [33, 35, 36].

Methodology

Our goal is to provide an end-to-end, fully unsupervised framework that assigns reliable fraud labels to highly imbalanced dataset without manual annotation or extensive domain expertise. We first apply unsupervised SHAP-based feature selection to identify the most informative feature which also reduces computational complexity. We then utilize our improved unsupervised labeling method [23]: (i) an ensemble of three unsupervised learners, which are implemented using scikit-learn's library [45] and (ii) a percentile-gradient. Their intersection yields a subset of confident labels (CL), which are used to create confidence intervals. The confidence intervals determine which positive instances will be kept in the final labeled dataset. We evaluate our labels directly and compare them against an unsupervised baseline.

Preprocessing the dataset

The actual labels are removed from the datasets prior to preprocessing; however, they are used, after labeling is complete, to evaluate the quality of the labels generated by our labeling method, which is explained in "Label evaluation methodology" section. First, we normalize the data using scikit-learn's normalize function [46], which scales each feature so that its Euclidean norm (L2 norm) equals 1 and ensures all features contribute equally, preventing features with larger scales from disproportionately impacting the results [47].

Following normalization, the dataset is randomized by shuffling the rows using pandas' sample method [48] with $frac = 1$. A random state (42) is applied, so that the randomized order is consistent across trial runs. Furthermore, we divide the data into smaller chunks of 25,000 rows to manage computational resources, given that the use of our labeling method is with large and big data. The total number of chunks is determined by dividing the dataset size by the chunk size and rounding up to ensure all rows are

included. Chunking is particularly useful when managing large and big datasets, as it allows for sequential processing of smaller portions, which reduces memory consumption, and improves computational efficiency. Given that our method employs three unsupervised algorithms, chunking provides efficiency and scalability for big data, by processing data in manageable portions [49]. This allows for large scale datasets, such as the Credit Card Fraud Detection dataset and Medicare Part D datasets, to be processed efficiently.

Ensemble unsupervised label generation method (EUM)

An initial set of labels are generated, to gain insight about the unlabeled dataset. Using the normalized and chunked datasets from “[Preprocessing the dataset](#)” section, three unsupervised algorithms: KMeans, Spectral Clustering (SC), and Gaussian Mixture Model (GMM) are applied. This combination of unsupervised algorithms has been demonstrated to be effective for labeling [22, 23].

KMeans partitions a dataset into k -clusters by iteratively minimizing the sum of squared distances between points and their cluster centroid. While KMeans is widely used for its simplicity and computational efficiency, it assumes spherical clusters, which can limit its effectiveness on complex or non-linearly separable data, thereby causing the algorithm to potentially misinterpret the actual cluster configuration [50].

Additionally, we use Spectral Clustering (SC) with a Nystroem Approximation for its effectiveness with large-scale datasets [51]. SC clusters data by partitioning a similarity graph based on the Laplacian eigenvectors, but constructing and decomposing the similarity matrix (or adjacency matrix) can be computationally expensive for large datasets. Because of this, we use the Nystroem approximation, which addresses this by approximating the graph eigenvectors using a subset of the data and reducing computational cost while maintaining clustering outcomes [51].

Gaussian Mixture Models (GMM) provide a probabilistic approach to clustering by representing data as a mixture of multiple Gaussian distributions. Each cluster is modeled as a distinct Gaussian component, characterized by its own mean and covariance matrix. The Expectation-Maximization (EM) algorithm is employed to iteratively optimize these parameters while assigning a probability for each data point’s membership in a given cluster. This soft clustering method allows GMM to effectively capture complex structures and overlapping clusters, making it particularly useful for identifying patterns in data [52].

An ensemble approach allows us to overcome limitations that each unsupervised algorithm may have, so that we are able to explore patterns and cluster structures in big data effectively. By forcing clusters to two, we align with the requirements of binary classification, which gives a simple assignment of labels to each instance and supporting our objective of binary labeling. While these algorithms are not traditionally used for labeling directly, our ensemble approach adapts their capabilities to effectively achieve this goal. For each algorithm (KMeans, SC, and GMM), we set `n_clusters` to two and `random_state` to zero. For KMeans, `n_init` is set to “auto”, and for SC, `affinity` is set to “Nearest Neighbor”. Spectral Clustering often struggles with large datasets due to the high computations and memory demands; however, we are able to overcome this limitation by applying the Nystroem method, which approximates large positive semi-definite

matrices by selecting a representative subset of data points [53], and scikit-learn's implementation controls the approximation's dimensionality via the default value for `n_components` [54].

In a post-processing step, the outputs of the algorithms are combined to create a label matrix, as shown in Fig. 1, which records the labels given by KMeans, SC and, and GMM for each instance. A majority vote is applied to determine a single label for each instance. Figure 1, shows an example label matrix along with its corresponding predicted label for each instance. To be able to map the clustering results to binary class labels, we assume that the minority cluster corresponds to the positive class (1) of the dataset. This assumption ensures that the labeling aligns with the dataset's characteristics and supports the subsequent labeling tasks.

Percentile gradient label generation method (PGM)

Using the output of the EUM labeling strategy, a calculation is taken to quantify the difference in the medians between the positive and negative class, for each feature. Given that the positive class is the class of interest, we focus on whether, generally, the positive class has larger or smaller values than the negative class. This calculation is detailed in “[Instance minimization for EUM and PGM](#)” section. From this determination, PGM generates a labeled dataset based on statistical thresholds and majority voting.

Using the normalized dataset, a binary label matrix is constructed to capture feature-based patterns within the data. Binary labels are assigned to each column based on statistical thresholds. Specifically, when higher feature values are expected for the positive class, the 97th percentile serves as the threshold, and data points exceeding this value are labeled as 1. Conversely, when lower feature values indicate positive instances, the 3rd percentile is used, assigning a 1 to data points below this threshold. All remaining data points are labeled as 0. This process results in a matrix, where each column corresponds to a feature, and each row represents a unique combination of binary labels, as detailed in Fig. 2.

A majority-voting approach is applied to each instance within the label matrix to ensure that each instance receives a single binary label. Determination of positive or negative is based on a threshold, set as a fixed fraction (0.009) of the total number of features. While this threshold is adjustable, experimentation has demonstrated that 0.009 yields effective results. Instances with a count of 1 s exceeding this threshold are assigned a final label of 1, whereas all others receive a label of 0. Essentially, this threshold defines the minimum number of positive labels (1 s) required for an instance to be labeled as positive.

$$\begin{bmatrix} \text{KMeans} & \text{SC} & \text{GMM} \\ 0 & 1 & 0 \\ \vdots & \vdots & \vdots \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} \text{Predicted Label} \\ 0 \\ \vdots \\ 1 \\ 0 \end{bmatrix}$$

Fig. 1 Label matrix example with predicted labels

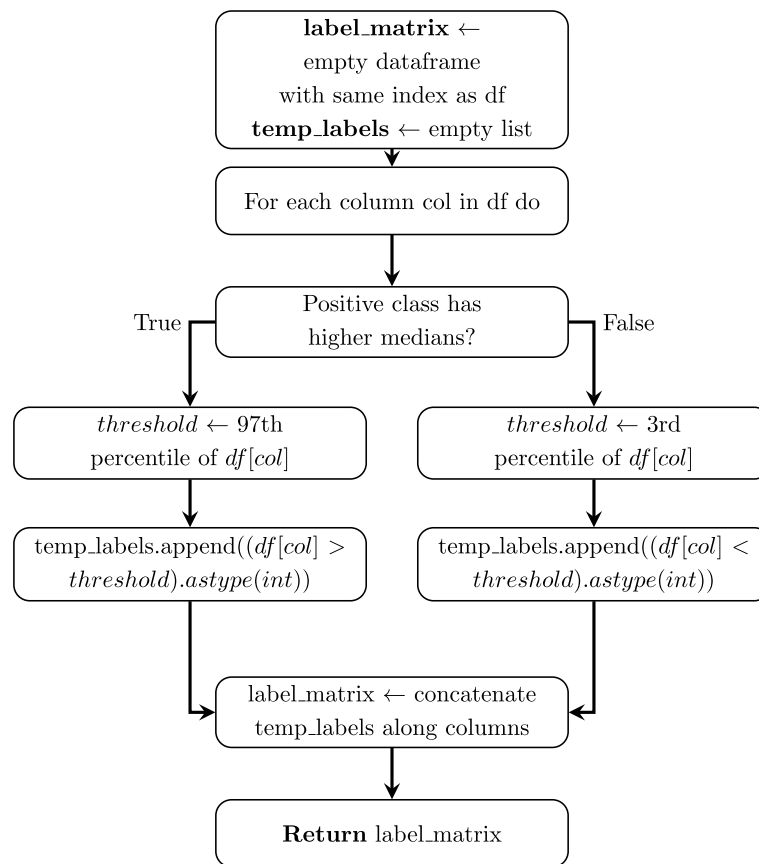


Fig. 2 Percentile gradient label matrix creation for imbalanced data

The majority vote process first computes the integer value of the total number of features multiplied by 0.009. For each row in the label matrix, the number of 1s is counted. If this count surpasses the computed threshold, the row is assigned a label of 1; otherwise, it is labeled as 0. This procedure results in a fully labeled dataset, where each instance is labeled based on voting across features.

Instance minimization for EUM and PGM

The focus of instance minimization is to minimize the number of positive instances, the class of interest, while capturing the highest number of true positive (actual fraud) instances. The process of instance minimization is applied to both the labeled datasets produced by EUM, “[Ensemble unsupervised label generation method \(EUM\)](#)” section, and PGM, “[Percentile gradient label generation method \(PGM\)](#)” section.

Using the initial labels generated by the EUM technique, as described in “[Ensemble unsupervised label generation method \(EUM\)](#)” section, we analyze feature distributions across the positive and negative classes by comparing their median values. Since the positive class represents fraudulent instances in our dataset, it serves as the primary focus of this analysis. This same class comparison approach is noted in “[Percentile gradient label generation method \(PGM\)](#)” section. To perform the analysis, the dataset is first separated by the class labels generated from EUM, and the median

for each feature is computed, excluding the class label column. The medians are then compared to determine the number of features where one class exhibits a higher median than the other. By counting the features where the positive class has a greater median value, we assess which class, generally, has higher feature values. Additionally, a Boolean indicator, *class_1_is_larger*, referenced in Algorithm 1, is assigned to indicate whether the positive class has more features with higher median values than the negative class. This determination plays an important role in the minimization process, particularly in refining which positive instances are kept in the final labeling.

The minimization technique, detailed in Algorithm 1, is applied to both EUM and PGM aiming to retain the correctly labeled true positives while correcting false negatives, or instances originally misclassified as 0. The assignment of *class_1_is_larger* influences how instances are relabeled, as outlined in Step 6 of Algorithm 1. If *class_1_is_larger* is false, the top-ranked 95% of rows labeled as 0 remain unchanged, while the bottom-ranked 5% are relabeled as 1. Among the rows labeled as 1, the top-ranked 1% retain their label, while the remaining 99% are relabeled as 0. If *class_1_is_larger* is true, the reverse process is applied. The bottom-ranked 95% of rows labeled as 0 remain unchanged, while the top-ranked 5% are relabeled as 1. Among the rows labeled as 1, the bottom-ranked 1% retain their label, while the remaining are relabeled as 0. This iterative relabeling process reduces false positives while preserving high-confidence positive instances, thereby enhancing the overall quality of labels generated by EUM and PGM.

Algorithm 1 Positive label minimization for EUM and PGM

```

1: Input: Dataframe with newly generated class label.
2: Output: Transformed dataset with updated labels.
3: Step 1: Sort Rows by Class
4: Sort the dataframe by the newly generated class label to group rows by labels (0 or 1).
5: Step 2: Split Data by Class
6:   Create two subsets:
7:     Class 0: Rows where newly generated label = 0.
8:     Class 1: Rows where newly generated label = 1.
9: Step 3: Compute Row Sums
10: For each subset, calculate the row-wise sum of feature values.
11: Step 4: Sort Rows by Row Sum
12:   Sort Class 0 in descending order of the row-wise sum.
13:   Sort Class 1 in ascending order of the row-wise sum.
14: Step 5: Define Thresholds for Relabeling
15:   Compute:
16:     Class 0 dataframe: Top 95% of rows.
17:     Class 0 dataframe: Bottom 5% of rows excluding the top 95%.
18:     Class 1 dataframe: Bottom 1% of rows.
19: Step 6: Relabel Rows
20:   If class_1_is_larger was determined to be false
21:     Keep highest-ranked 95% of rows in Class 0 dataframe unchanged.
22:     Relabel lowest-ranked 5% of rows in Class 0 dataframe as 1.
23:     Select the rest of the rows in Class 0 dataframe and keep label as 0.
24:     Keep highest-ranked 1% of rows in Class 1 dataframe unchanged.
25:     Relabel remaining rows in Class 1 dataframe as 0.
26:   If class_1_is_larger was determined to be true
27:     Keep lowest-ranked 95% of rows in Class 0 dataframe unchanged.
28:     Relabels highest-ranked 5% of rows in Class 0 dataframe as 1.
29:     Select the rest of the rows in Class 0 dataframe and keep label as 0.
30:     Keep lowest-ranked 1% of rows in Class 1 dataframe unchanged.
31:     Relabel remaining rows in Class 1 dataframe as 0.
32: Step 7: Combine Transformed Data
33:   Concatenate all transformed subsets into a single dataframe.

```

Confident label creation

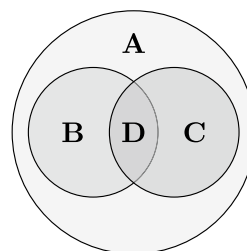
From the minimized datasets produced from EUM and PGM, mEUM and mPGM, respectively, a subset is created called the Confident Labels (CL). The CLs are created from the intersection of instances present in both mEUM and mPGM which share the same predicted label. This is illustrated in Fig. 3. The CL subset captures instances to help determine the positive instances which will be kept in the final labeling, as outlined in “Controlling positive instances” section.

Controlling positive instances

The final step in the labeling process involves determining the desired number of positive instances, p . Since negative instances are more predictable due to their majority presence in the dataset, the final adjustment focuses primarily on defining the count of positive instances. Once p is specified, the process first checks whether the minimized PGM method (mPGM) has a sufficient number of positive instances. If enough positives exist, only those instances are considered for the final positive labeling, and any remaining unused instances are relabeled as negative. However, if mPGM does not contain enough positive instances to meet p , additional negative instances are selected and relabeled as positive to reach the required count, as described below.

To refine the selection process, confidence intervals are constructed for each feature based on the positive instances using the CLs described in “Confident label creation” section. If p is less than or equal to the number of positives in the mPGM dataset, only the instances labeled as positive are considered. If p exceeds the available positives in mPGM, additional negative instances are evaluated for potential relabeling. The fully labeled mPGM dataset is then separated by class, and for each positive instance, feature values are checked against their respective confidence intervals. Instances with feature values falling within these intervals are labeled as positive, while those outside are assigned a negative label. Once this process is completed across all features and instances, the total count of features falling within confidence intervals is determined for each instance. Instances are then sorted based on these counts, and rows receive a final label of positive until the desired number of positives, p , is reached. Any remaining unused rows are relabeled as negative, completing the final labeling of the dataset.

In cases where p exceeds the available positives in the labeled mPGM dataset, all positive instances are retained, and additional negative instances are selected to fulfill



Key:

- A: Original Unlabeled Dataset
- B: Minimized Ensemble Unsupervised (mEUM) (Section 3.2)
- C: Minimized Percentile-Gradient (mPGM) (Section 3.3)
- D: Confident Labels Subset (CL) (Section 3.5)

Fig. 3 Venn diagram demonstrating confident label creation

the required count. Negative instances are ranked using the same feature-based confidence interval approach, and the top-ranked instances are relabeled as positive until p is reached. Any remaining instances retain their label of negative. This process ensures that the dataset is fully labeled, and finalizes the assignment of class labels to every instance.

Label evaluation methodology

Traditionally, newly generated binary class labels are evaluated indirectly by training a supervised model on the generated labels and measuring the model's performance. However, this approach presents challenges, as variations in classifiers and datasets can lead to inconsistent results due to biases and interactions inherent to different modeling techniques [44]. To overcome these limitations, this paper introduces a direct label evaluation approach that assesses label quality without relying on a supervised model. Instead of measuring how well a classifier performs on the generated labels, the proposed method evaluates label quality directly using metrics such as Precision, Recall, F1-score, Jaccard Index (JI), and Matthews Correlation Coefficient (MCC), which are derived from true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN).

By focusing on direct evaluation, this approach eliminates biases introduced by dataset-classifier interactions, resulting in a more consistent and clear assessment of label quality. The generated labels are compared against the ground truth labels to calculate the evaluation metrics. Ground truth labels are used solely for evaluation for both our labeling method and the baseline. To evaluate the overall quality of the labels generated by our method, we compare them against those produced by an unsupervised baseline. However, to the best of our knowledge, no fully automated and entirely unsupervised labeling method currently exists. Therefore, we selected Isolation Forest (IF) [10] as our unsupervised baseline because of its suitability for large datasets. Among state-of-the-art (SOTA) unsupervised methods, IF has demonstrated performance comparable to alternatives such as One-Class SVM (OCSVM) and Local Outlier Factor (LOF) [41]. However, unlike these methods, which may suffer from scalability issues due to their computational complexity, IF maintains high accuracy while offering significantly lower computational complexity, making it a suitable selection for large and big data [12, 55]. We assess the label quality generated by our methodology against the label quality generated by IF. Our empirical results demonstrate that our labeling method outperforms IF, producing higher quality labels. Therefore, our method provides a robust unsupervised labeling method suitable for real-world scenarios for large and big data, when labeled data is unavailable.

Experimental design

The aim of our experiments is to evaluate both the quality of our method's newly generated binary class labels, as well as the impact to the quality of labels produced when features are minimized using unsupervised SHAP-based feature selection on severely imbalanced fraud data. We applied this method to the Credit Card Fraud Detection dataset [5] and Medicare Part D dataset [6]. The methodology is outlined here, with dataset-specific details provided in "[Credit card fraud detection dataset](#)" section for the Credit Card dataset and "[Medicare part D dataset](#)" section for Medicare Part D.

First, we applied unsupervised SHAP-based feature selection to rank feature importance for each dataset. For the Credit Card dataset, we selected the top three and five features in addition to using the full-feature dataset (29 features). Similarly, for the Medicare Part D dataset, we selected the top three and seven features along with the full-featured dataset (82 features). This results in a total of six data subsets. Across both datasets, these feature combinations, three and five features for Credit Card, and three and seven features for Medicare Part D, consistently produced the highest evaluation metrics when compared to both the baseline and the full-featured dataset.

Using our unsupervised labeling method, we generated binary class labels for each dataset, both with and without feature selection, and evaluated the quality of the generated labels. Additionally, we apply an unsupervised baseline method to generate labels to each dataset, with and without feature selection, and compared their label quality against our method. Notably, class labels were generated exclusively from features, and the ground truth labels were used solely for evaluation purposes. Our methodology initially assigns positive and negative labels, which are then refined to reduce the number of positives (fraudulent instances), ensuring that only those with the highest confidence in representing actual fraud are retained and returns a fully labeled dataset.

Case study datasets

The datasets used in our experiments are both publicly available and widely utilized in fraud detection research. The first dataset [5] comprises over 280,000 European credit card transactions from September 2013. To the best of our knowledge, this is the only publicly available dataset of its kind. Additionally, we employ a Medicare Part D dataset spanning 2013 to 2019, which provides detailed records of prescription drugs dispensed to Medicare beneficiaries [6]. Both datasets exhibit severe class imbalance and are derived from real-world financial and healthcare data, making them highly relevant for evaluating fraud detection methodologies.

Credit card fraud detection dataset

As previously mentioned, the Credit Card Fraud Detection dataset [5] is widely used in fraud detection research. This dataset was developed through a collaboration between Worldline, a leading payment processing company, and the Université Libre de Bruxelles, with over 280,000 credit card transactions. Each transaction is represented by 30 anonymized features (“V1” through “V28”, “Time”, and “Amount”). The “V” features were transformed using Principal Component Analysis (PCA) by the original dataset creators to protect sensitive information.

As is common in fraud detection, the dataset exhibits severe class imbalance, with legitimate transactions (negative class) significantly outnumbering fraudulent transactions (positive class), with the positive being the class of interest. The dataset’s characteristics are summarized in Table 1. There are three datasets, two created from the top features selected by SHAP, Credit Card Fraud—3 Features and Credit Card Fraud—5 Features, plus the original full-featured dataset with 29 features. While the number of features varies, the total instances and imbalance remain the same. Although this dataset includes labels, our unsupervised labeling methodology does not utilize these labels except to evaluate the labels generated by our method. During the labeling process, the

Table 1 Credit card fraud detection dataset class characteristics

Dataset	Majority count	Minority count	Total count	Minority imbalance	Features
Credit card fraud	284,315	492	284,807	0.1727%	29
Credit card fraud—3 features	284,315	492	284,807	0.1727%	3
Credit card fraud—5 features	284,315	492	284,807	0.1727%	5

Table 2 Medicare part D dataset class characteristics

Dataset	Majority count	Minority count	Total count	Minority imbalance	Features
Medicare part D	5,340,406	3700	5,344,106	0.0692%	82
Medicare part D—3 features	5,340,406	3700	5,344,106	0.0692%	3
Medicare part D—7 features	5,340,406	3700	5,344,106	0.0692%	7

original class labels are removed, ensuring that our method remains fully unsupervised. Furthermore, the “Time” feature is excluded, as prior to labeling as it has shown that it lacks meaningful predictive value [56].

Medicare part D dataset

In addition, we utilize the publicly available Medicare Part D dataset spanning 2013 to 2019 [57]. Administered by CMS, this dataset provides detailed records on prescription drugs dispensed to Medicare beneficiaries enrolled in Prescription Drug Coverage, with various attributes such as drug prescription records, prescribing patterns, and provider details. Each healthcare provider is uniquely identified by their National Provider Identifier (NPI), and the dataset aggregates records across NPIs, years, and prescriber types. While it originally only contained 31 features, it was expanded to 82 features by integration of supplementary provider data from the CMS Chronic Conditions Data Warehouse.

The challenge was that the Medicare Part D dataset from CMS does not include explicit fraud labels. To address this, fraudulent providers were identified using the List of Excluded Individuals and Entities (LEIE) dataset from the U.S. Department of Health and Human Services [58]. Updated monthly, the LEIE, catalogs individuals and entities excluded from federally funded healthcare programs due to violations from prohibited activities. Fraudulent providers were identified by matching NPIs from the LEIE dataset to those in the Medicare Part D dataset. Following the methodology outlined in [59], providers excluded for fraudulent activities were labeled accordingly.

This process resulted in the identification of 3700 fraudulent providers, creating a dataset with a severe class imbalance. Legitimate providers dramatically outnumber fraudulent ones. The methodology used to integrate these fraud labels and enhance the dataset is further detailed in [6]. A summary of the fully compiled and labeled Medicare Part D dataset is provided in Table 2, which also includes the addition of the two data subsets created by minimizing the features from the selection of the top three and seven features, titled Medicare Part D—3 Features and Medicare Part D—7 Features, respectively, from SHAP.

Evaluation methodology

To assess the quality of the labels generated by our unsupervised method, we employ five widely recognized evaluation metrics: Matthew's Correlation Coefficient (MCC), Jaccard Index (JI), Precision, Recall, and F1-score. These metrics provide a comprehensive evaluation of label quality, particularly in the context of highly imbalanced datasets [38, 60, 61]. When used in conjunction, these metrics demonstrate a clear way to evaluate the quality of the generated labels from our labeling method.

Matthew's Correlation Coefficient (MCC), defined in Eq. 1, provides a balanced evaluation of binary labels as it considers all four elements of a confusion matrix. Unlike accuracy, MCC is appropriate when severe class imbalance is present; therefore, providing a reliable assessment of label quality [4, 38, 61].

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \quad (1)$$

Jaccard Index (JI), also known as Intersection over Union, defined in Eq. 2, is similar to Precision but penalizes for both false positives and false negatives whereas Precision only penalizes for false positives. JI provides a straight-forward measurement of how well the predicted positive instances align with the actual positive instances, as well as accounting for both false positives and false negatives [39]. JI is useful for imbalanced datasets, as it emphasizes positive class predictions and ensures that models are evaluated based on their ability to correctly identify fraudulent instances [39, 60].

$$\text{Jaccard Index} = \frac{TP}{TP + FP + FN} \quad (2)$$

Precision, defined in Eq. 3, measures the proportion of correctly identified positive labels among all predicted positives. It is a highly relevant metric when false positives carry significant cost, as is the case in fraud detection. In fraud detection, false positives can result in wasted resources, unnecessary investigations, and loss of trust. High Precision ensures that instances marked as fraudulent are likely to be genuine fraud cases, which optimizes efficiency and minimizes unnecessary expenses [38].

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

Recall, defined in Eq. 4, also measures the proportion of correctly identified positive labels; however, it penalizes for false negatives rather than false positives, like Precision. Missing fraudulent transactions, in other words marking a fraudulent transaction as legitimate (false negative), can lead to financial loss and undetected fraudulent activity. High value for recall ensures that the majority of fraudulent cases were identified [38, 62].

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

F1-score, defined in Eq. 5, represents the harmonic mean of Precision and Recall, balancing the trade-off between false positives and false negatives. It ranges from 0 to 1, with higher values indicating better overall performance. This metric is particularly

valuable when the data is imbalanced, since relying solely on Precision or Recall may provide a misleading assessment of model effectiveness [63].

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

By incorporating all five evaluation metrics, we conduct a comprehensive assessment of the quality of labels produced by our unsupervised labeling method. Precision ensures that detected fraudulent instances are highly likely to be actual fraud cases, while Recall verifies that a substantial portion of fraudulent cases are correctly identified. F1-score and MCC provide a balanced evaluation of classification performance, accounting for both false positives and false negatives. Meanwhile, the Jaccard Index quantifies the overlap between predicted and ground-truth labels. Collectively, these metrics offer a holistic evaluation of label quality, ensuring that the generated labels are both accurate and reliable for further analysis and decision-making.

Baseline comparison

To evaluate the impact of SHAP-based feature selection on binary label generation within our labeling method, we compared the labels produced by our approach to those generated by an unsupervised baseline, the Isolation Forest (IF). Isolation Forest, developed by Liu et al. [10], is a widely recognized unsupervised anomaly detection method well-suited for handling large-scale datasets. IF has demonstrated superior performance compared to other unsupervised learning techniques [11], offering high accuracy with low computational complexity [12], and avoiding the unpredictability associated with clustering-based anomaly detection [13]. We used the scikit-learn implementation of IF [45] and set the contamination rate to “auto”, its default value. This comparison was done across SHAP-selected datasets containing 3 and 5 features as well as the full-featured dataset, for the Credit Card Fraud Detection dataset, and the datasets containing 3 and 7 features plus the full-featured dataset, for the Medicare Part D dataset.

The IF algorithm detects anomalies by constructing an ensemble of binary decision trees, where each tree is built using randomly selected features. Data points that require shorter average path lengths for isolation are more likely to be anomalies. IF is particularly advantageous due to its efficiency in high-dimensional spaces and resilience to overfitting. Additionally, its computational scalability makes it an appropriate selection for large and big datasets.

Results and analysis

Our experiments are designed to evaluate the impact of unsupervised SHAP-based feature selection within our novel unsupervised labeling method. By assessing label quality directly, we are able to clearly see the positive impact of SHAP when integrated with our unsupervised labeling method. Additionally, the reduction of features allows for simplified analysis and reduces computational complexity, given we are experimenting with large and big data. We present five evaluation metrics for both datasets in the tables below. Our selection of evaluation metrics have been studied for their appropriateness in use with large and big data, across varying levels of positives for each dataset [38, 60, 61]. Our labeling method was applied to both the Credit Card Fraud Detection dataset

Table 3 Metrics across feature subsets and positive class counts—credit card

Features	Positive	MCC	Jaccard index	Precision	Recall	F1-score
3	500	0.3254	0.1952	0.3240	0.3293	0.3254
	600	0.3356	0.2013	0.3050	0.3720	0.3352
	700	0.3394	0.2016	0.2857	0.4065	0.3356
	800	0.3397	0.1985	0.2675	0.4350	0.3313
	900	0.3351	0.1918	0.2489	0.4553	0.3218
	1000	0.3334	0.1870	0.2350	0.4776	0.3150
5	500	0.3780	0.2338	0.3760	0.3821	0.3790
	600	0.3761	0.2311	0.3417	0.4167	0.3761
	700	0.3685	0.2226	0.3100	0.4411	0.3685
	800	0.3668	0.2177	0.2888	0.4695	0.3668
	900	0.3622	0.2104	0.2689	0.4919	0.3477
	1000	0.3535	0.2003	0.2490	0.5061	0.3338
29	500	0.2810	0.1643	0.2800	0.2846	0.2823
	600	0.2618	0.1507	0.2383	0.2907	0.2619
	700	0.2489	0.1407	0.2100	0.2988	0.2466
	800	0.2726	0.1536	0.2150	0.3496	0.2663
	900	0.2959	0.1658	0.2200	0.4024	0.2845
	1000	0.3077	0.1702	0.2170	0.4411	0.2909

Table 4 Baseline isolation forest (IF) metrics—credit card

Features	MCC	Jaccard index	Precision	Recall	F1-score
3	0.0435	0.0068	0.0068	0.4411	0.0134
5	0.1009	0.0147	0.0147	0.8028	0.0289
29	0.1751	0.0385	0.0388	0.8333	0.0741

with and without feature selection, see “[Credit card dataset results and analysis](#)” section, and the Medicare Part D dataset with and without feature selection, see “[Medicare part D dataset results and analysis](#)” section.

Credit card dataset results and analysis

Table 3 presents the results of our experiment on the Credit Card Fraud Detection dataset using our labeling method with and without feature selection using the evaluation metrics of Matthew’s Correlation Coefficient (MCC), Jaccard Index (JI), Precision, Recall, and F1-score. The corresponding results of applying our baseline unsupervised method, Isolation Forest (IF), are presented in Table 4.

Using the original, full-featured dataset, MCC reaches the peak value at 1000 total positive instances. MCC diminishes from 500 to 700 and increases from 700 to 1000. JI, follows a similar pattern, with decreases from 500 to 700 and increasing from 700 to 1000, with the highest metric value at 1000. Precision, on the other hand, reaches the peak value at 500 and diminishes between 500 and 700 with a slight increase from 700 to 900 and slight decrease from 900 to 1000. Lastly, Recall, has its peak value at 1000 and least value at 500. Our method results in mean values are 0.2779, 0.1575, 0.2301, 0.3445, and 0.2332 for MCC, JI, Precision, Recall, and F1-score, respectively. However,

for IF, the evaluation metrics are 0.1751, 0.0385, 0.0388, and 0.8333 for MCC, JI, Precision, and Recall, respectively. Our labeling method has notably higher values for MCC, JI, and Precision and lower value for Recall. Given that MCC is a metric that considers all parts of the confusion matrix, it is a helpful wholistic value to determine overall quality of the labels, which our method ranged from the lowest value at 0.2489 to the highest value at 0.3077 while IF has a value of 0.1751 demonstrating our unsupervised labeling method produced higher quality labels overall. Precision heavily penalizes for false positive instances, or in our case legitimate transactions labeled as fraudulent, which means resources will be wasted investing legitimate transactions. Our method has a minimum value of 0.2100 and maximum value at 0.2800, for Precision, whereas IF has a Precision value of 0.0388. JI considers both false positives (legitimate labeled as fraud) and false negatives (fraudulent labeled as legitimate). Again, similar to MCC, JI is a more wholistic view of labeling. In the case of JI, our method has a minimum value of 0.1407 and maximum value of 0.1702, while IF has a JI value of 0.0385. Thus, this demonstrates our method has higher quality labels overall. Lastly, we consider Recall. IF outperforms our method with a value of 0.8333 and our method having a minimum of 0.2846 and maximum value of 0.4411. However, it is important to consider the balance of Precision and Recall, given they penalize for different things. The way to determine which method produces higher quality labels overall is to consider F1-score, which balances Precision and Recall. F1-score for IF is 0.0741 whereas F1-score for our labeling method ranges from a minimum of 0.2466 to a maximum of 0.2909, a significant improvement over the baseline. Therefore, this demonstrates, overall, our method provides higher quality labels.

For the dataset with the top three features, MCC is relatively stable between 500 and 1000 positive instances, with a slight increase from 500 to 800, followed by a minor decline from 800 to 1000; however, the MCC value at 1000 is slightly higher than at 500. The addition of positive instances contributes to the increase in true positives. Similarly, JI is fairly stable between 500 and 1000 instances, with a slight increase from 500 to 700 and slight decrease from 700 to 1000. However, for JI, the value at 1000 is slightly less than the value at 500. Precision reaches its highest value at 500, followed by a decline from 500 to 1000. This trend occurs because the number of positives identified at 500 is high, and although the number of true positives increases as additional positives are introduced, they are added at a slower rate relative to the overall increase in positive instances. On the other hand, Recall, reaches its highest value at 1000, with its lowest value at 500. IF, using only the top three features, has values of 0.0435, 0.0068, 0.0068, and 0.4411, for MCC, JI, Precision, and Recall, respectively. In comparison, the application of our unsupervised labeling method resulted in mean values for MCC, JI, Precision, and Recall of 0.3348, 0.1959, 0.2777, and 0.4126, respectively. There is a significant improvement for MCC, JI, and Precision; however, Recall is slightly less than the baseline. Once more, we will consider F1-score for both our labeling method and the baseline. Our labeling results in a mean F1-score of 0.3274, with a minimum value of 0.3150 and a maximum value of 0.3356, while IF results in a F1-score of 0.0134. Thus, this demonstrates our labeling method outperforms the baseline overall. Additionally, when comparing three features to the full-featured metric results, we see the mean values for MCC, JI, Precision, Recall, and F1-score are 0.2779, 0.1575, 0.2301, 0.3445, and 0.2720, respectively, whereas using only 3 features results in mean values for MCC, JI, Precision,

Recall, and F1-score of 0.3348, 0.1959, 0.2777, 0.4126, and 0.3274, respectively, clearly demonstrating significant improvement when compared to the full-featured dataset.

Our methods labeling for the top 5 five features, for MCC, has the highest value at 500 with a slight diminishment in value from 500 to 1000; however, the lowest value of MCC for 5 features is higher than MCC for 3 features. Similarly, to MCC, JI has the largest value at 500 and diminishes from 500 to 1000. Precision also follows the trend with the highest value at 500 and diminishing from 500 to 1000. Lastly, Recall has the lowest value at 500 and steadily increases between 500 and 1000.

IF, using only the top five features, has values of 0.1009, 0.0147, 0.0147, and 0.8028, for MCC, JI, Precision, and Recall, respectively. In comparison, the application of our unsupervised labeling method resulted in mean values for MCC, JI, Precision, and Recall of 0.3675, 0.2193, 0.3057, and 0.4512, respectively. Our unsupervised method clearly outperforms the baseline for MCC, JI, and Precision; however, for Recall, similarly to the full-featured dataset, our method has a lower value. Again, we will consider the F1-score as a wholistic determination of the trade-off of Precision and Recall. Our method results in an average F1 value of 0.3620, with a minimum value of 0.3338 and a maximum value of 0.3790, whereas, IF results in an F1 score of only 0.0289. Thus, this reaffirms our methods ability to outperform the baseline.

Additionally, when we compare the mean values for each metric for three and five features, we find that five features has higher values for each metrics. Three features results in mean values of 0.3348, 0.1959, 0.2777, 0.4126, and 0.3274 for MCC, JI, Precision, Recall, and F1-score while five features results in mean values of 0.3675, 0.2193, 0.3057, 0.4512, and 0.3620 for MCC, JI, Precision, Recall, and F1-score. Lastly, we compare the mean values for each metrics for five features compared to the mean values for the full-featured dataset and see that there is a significant improvement with the full-featured dataset having mean values of 0.2779, 0.1575, 0.2301, 0.3445, and 0.2720 and five features have mean values of 0.3675, 0.2193, 0.3057, 0.4512, and 0.3620 for MCC, JI, Precision, Recall, and F1-score, respectively. Consequently, we determine that unsupervised SHAP-based feature selection improves our unsupervised labeling method.

Medicare part D dataset results and analysis

Table 5 presents the results of our experiment on the Medicare Part D dataset with and without feature selection for our labeling method and shows the results of the calculation of our evaluation metrics: Matthew's Correlation Coefficient (MCC), Jaccard Index (JI), Precision, Recall, and F1-score. The corresponding results of applying our baseline unsupervised method, Isolation Forest (IF), are presented in Table 6.

The full-featured dataset, which has 82 features, results in a peak MCC value of 0.5460 and minimum value of 0.1182. From 1000 to 3000, there is a significant increase with a slight decrease from 3000 to 5000. Again, there is a significant jump from 5000 to 7000 and a slight diminishment from 7000 to 13,000. JI follows the same pattern of increases and decreases as MCC; however, the peak value for JI is 0.3512 with a minimum value of 0.0510. Precision, on the other hand, has a peak value at 3000 of 0.5030. It increases from 1000 to 3000, decrease from 3000 to 5000, increases again from 5000 to 7000, and ultimately diminishes from 7000 to 13,000. Lastly, Recall has a peak value of 0.7965 at 13,000. Recall steadily increases from 1000 to 13,000 positive values. F1-score has a peak

Table 5 Metrics across feature subsets and positive class counts—medicare part D

Features	Positive	MCC	Jaccard index	Precision	Recall	F1-score
3	1000	0.5197	0.2703	1.0000	0.2703	0.4255
	3000	0.4655	0.3015	0.5173	0.4195	0.4633
	5000	0.3603	0.2171	0.3104	0.4195	0.3568
	7000	0.3043	0.1697	0.2217	0.4195	0.2901
	9000	0.3310	0.1775	0.2127	0.5173	0.3014
	11,000	0.5766	0.3341	0.3346	0.9949	0.5008
	13,000	0.5330	0.2846	0.2846	1.0000	0.4431
7	1000	0.5197	0.2703	1.0000	0.2703	0.4255
	3000	0.4352	0.2764	0.4837	0.3922	0.4331
	5000	0.3368	0.2002	0.2902	0.3922	0.3336
	7000	0.2844	0.1569	0.2073	0.3922	0.2712
	9000	0.4779	0.2778	0.3068	0.7462	0.4348
	11,000	0.5687	0.3280	0.3301	0.9814	0.4940
	13,000	0.5330	0.2846	0.2846	1.0000	0.4431
82	1000	0.1182	0.0510	0.2280	0.0616	0.0970
	3000	0.4526	0.2907	0.5030	0.4078	0.4504
	5000	0.3503	0.2098	0.3018	0.4078	0.3469
	7000	0.5460	0.3512	0.3973	0.7516	0.5198
	9000	0.4989	0.2935	0.3202	0.7789	0.4539
	11,000	0.4609	0.2504	0.2676	0.7957	0.4005
	13,000	0.4243	0.2143	0.2267	0.7965	0.3529

Table 6 Baseline isolation forest (IF) metrics—medicare part D

Features	MCC	Jaccard index	Precision	Recall	F1-score
3	0.0047	0.0010	0.0010	0.2070	0.0020
7	0.0077	0.0013	0.0013	0.2008	0.0025
82	0.0156	0.0021	0.0021	0.2362	0.0042

value of 0.5198 at 7000 positives. F1-score increases from 1000 to 3000 positives with a decrease from 3000 to 5000 and another increase from 5000 to 7000 with a final diminishment from 7000 to 13,000.

When comparing the full-featured dataset labels to our baseline, IF, there is a significant increase in label quality. The results from our labeling method have mean values of 0.4073, 0.2372, 0.3206, 0.5714, and 0.3744 for MCC, JI, Precision, Recall, and F1-score, respectively, whereas IF has values of 0.0156, 0.0021, 0.0021, 0.2362, and 0.0042 for MCC, JI, Precision, Recall, and F1-score, respectively. Across all evaluation metrics, our labeling method is a significant improvement over the baseline.

Looking at the results from the dataset with only three features, the MCC values start at 0.5197 at a positive level of 1000 and decrease until 7000. From 7000 to 11,000 there is an increase with a slight diminish from 11,000 to 13,000. JI increases from 1000 to 3000, then declines from 3000 to 7000, increases again from 7000 to 11,000 and tappers off slightly again from 11,000 to 13,000. Precision and Recall both reach values of 1.000. Precision achieves that at the 1000 positive level, but then diminishes from 1000 to 13,000. Recall, on the other hand, starts with a value of 0.2703 and increases from 1000 to 13,000

until it reaches a value of 1.000, meaning at this point all fraudulent instances have been found. F1-score increases from 1000 to 3000, then declines from 3000 to 7000, increases again from 7000 to 11,000 and diminishes slightly again from 11,000 to 13,000.

Similarly, for seven features, the MCC values follow the same increase and decreases at three features, with values ranging from a minimum of 0.2844, at positive level 7000, to a maximum of 0.5687, at positive level 11,000. JI, also follows a similar pattern as three features with a minimum value of 0.1569, at positive level 7000, and maximum value of 0.3280, at positive level 11,000. For seven features, Precision also has a maximum value of 1.000, at positive level 1000, and minimum value of 0.2846, at the 13,000 level. Recall, started at a minimum value of 0.2703, at positive level 1000, and increased to reach a maximum value of 1.000, at positive level 13,000. F1-score increases from 1000 to 3000, then declines from 3000 to 7000, increases again from 7000 to 11,000 and tappers off slightly again from 11,000 to 13,000. For both three and seven features at positive level 1000 and 13,000, the values of MCC, JI, Precision, Recall, and F1-score are identical, though vary in the positive levels between. In other words, both data subsets are able to find all fraudulent instances.

However, when we compare the results of three features and seven features to the corresponding values for MCC, JI, Precision, Recall, and F1-score for the baseline, it is clear to distinguish the tremendous difference in metric values and the negative impact feature selection has on the baseline compared to the positive impact it has on our unsupervised labeling method. For three features, our labeling method has mean values of 0.4415, 0.2506, 0.4116, 0.5773, and 0.3972 for MCC, JI, Precision, Recall, and F1-score, respectively while IF has values of 0.0047, 0.0010, 0.0010, 0.2070, and 0.0020 for MCC, JI, Precision, Recall, and F1-score, respectively. For seven features, our labeling method has mean values of 0.4508, 0.2563, 0.4145, 0.5963, and 0.4050 for MCC, JI, Precision, Recall, and F1-score, respectively whereas IF has values of 0.0077, 0.0013, 0.0013, 0.2008, and 0.0025 for MCC, JI, Precision, Recall, and F1-score, respectively. Additionally, we want to compare the two datasets with feature selection to our original dataset without feature selection. Again, as evidenced by the empirical results in Table 5, both datasets with feature selection show improved results when compared to the full-featured dataset with the additional benefit of being less computationally complex as well as having easier interpretability given the significant minimization of features.

Conclusion

With the majority of new data being unlabeled and considering challenges such as class imbalance, privacy, and expense associated with manual annotation, the development of reliable unsupervised labeling techniques is needed. Through our study, we thoroughly evaluate the improvement to label quality when applying unsupervised SHAP-based feature selection in conjunction with our unsupervised labeling method for two severely imbalanced fraud datasets, namely Credit Card Fraud Detection dataset and the Medicare Part D dataset. Without the requirement of labels, we use SHAP feature selection to make four additional data subsets, with two for each dataset, Credit Card and Medicare Part D, by selecting varying numbers of the top features, as well as using the full-featured original version of the dataset. Each dataset with and without feature selection are then labeled. Our approach effectively labels

unlabeled, large and big data with severe class imbalance while reducing computational complexity by reducing the number of features needed to label, which also allows for easier interpretability given the reduction in features. This is all able to be done in a fully unsupervised framework.

Using MCC, JI, Precision, Recall, and F1-score, we evaluate the quality of the labels produced by our unsupervised labeling method, with and without feature selection, for both datasets. Comparing the results to the unsupervised baseline method, Isolation Forest (IF), we demonstrate SHAP-based feature selection significantly enhances label quality. For the Credit Card Fraud Detection dataset, the top 5 features selected by SHAP consistently produced the highest quality labels, as demonstrated by the increase in metric values, outperforming the labels generated by the 3 features, full-features and the baseline. For the Medicare Part D Dataset, the top 7 features consistently outperformed the 3 features, full-features, and the baseline. Additionally, for the Medicare Part D dataset, for both 3 and 7 features, all fraudulent instances were captured whereas for the full-featured dataset and the baseline, this was not true. Future work will continue to explore additional applications of our labeling method across different domains as well as the automation of the optimal number of positives for a given dataset.

Abbreviations

CI	Confidence interval
CL	Confident labels
EUM	Ensemble unsupervised method
GMM	Gaussian mixture model
FN	False negative
FP	False positive
IF	Isolation forest
JI	Jaccard index
MCC	Matthew's correlation coefficient
mEUM	Minimized ensemble unsupervised method
mPGM	Minimized percentile gradient method
PGM	Percentile gradient method
SC	Spectral clustering
TN	True negative
TP	True positive

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Author contributions

MAW carried out the concept and design of the research, performed the implementation and experimentation, and contributed to the manuscript. TMK contributed to the design of the research and provided reviews to the manuscript. All authors read and approved the final manuscript.

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No datasets were generated or analysed during the current study.

Declarations

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Competing interests

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